



Submission Deadline: 03/03/25	LA & Diff. Eqs (MT 102) Spring 2024	Assignment 2
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- 1. Powers of i .** Show that $i^2 = -1, i^3 = -i, i^4 = 1, i^5 = i, \dots$ and $1/i = -i, 1/i^2 = -1, 1/i^3 = i, \dots$.
- 2. Rotation.** Multiplication by i is geometrically a counterclockwise rotation through $\pi/2$ (90°). Verify

Let $z_1 = -2 + 11i, z_2 = 2 - i$. Showing the details of your work, find, in the form $x + iy$:

- 8.** $z_1 z_2, \overline{(z_1 z_2)}$
- 9.** $\operatorname{Re}(z_1^2), (\operatorname{Re} z_1)^2$
- 10.** $\operatorname{Re}(1/z_2^2), 1/\operatorname{Re}(z_2^2)$
- 11.** $(z_1 - z_2)^2/16, (z_1/4 - z_2/4)^2$
- 12.** $z_1/z_2, z_2/z_1$
- 13.** $(z_1 + z_2)(z_1 - z_2), z_1^2 - z_2^2$
- 14.** $\bar{z}_1/\bar{z}_2, \overline{(z_1/z_2)}$
- 15.** $4(z_1 + z_2)/(z_1 - z_2)$

Represent in polar form and graph in the complex plane as in Fig. 325. Do these problems very carefully because polar forms will be needed frequently. Show the details.

- | | |
|--|---|
| 1. $1 + i$ | 2. $-4 + 4i$ |
| 3. $2i, -2i$ | 4. -5 |
| 5. $\frac{\sqrt{2} + i/3}{-\sqrt{8} - 2i/3}$ | 6. $\frac{\sqrt{3} - 10i}{-\frac{1}{2}\sqrt{3} + 5i}$ |
| 7. $1 + \frac{1}{2}\pi i$ | 8. $\frac{-4 + 19i}{2 + 5i}$ |

9-14 PRINCIPAL ARGUMENT

Determine the principal value of the argument and graph it as in Fig. 325.

- | | |
|--------------------|----------------------------|
| 9. $-1 + i$ | 10. $-5, -5 - i, -5 + i$ |
| 11. $3 \pm 4i$ | 12. $-\pi - \pi i$ |
| 13. $(1 + i)^{20}$ | 14. $-1 + 0.1i, -1 - 0.1i$ |

15-18 CONVERSION TO $x + iy$

Graph in the complex plane and represent in the form $x + iy$:

- | | |
|---|---|
| 15. $3 (\cos \frac{1}{2}\pi - i \sin \frac{1}{2}\pi)$ | 16. $6 (\cos \frac{1}{3}\pi + i \sin \frac{1}{3}\pi)$ |
| 17. $\sqrt{8} (\cos \frac{1}{4}\pi + i \sin \frac{1}{4}\pi)$ | |
| 18. $\sqrt{50} (\cos \frac{3}{4}\pi + i \sin \frac{3}{4}\pi)$ | |

NOTE

Find the eigenvalues. Find the corresponding eigenvectors.
Use the given λ or factor in Probs. 11 and 15.

1. $\begin{bmatrix} 3.0 & 0 \\ 0 & -0.6 \end{bmatrix}$

2. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

3. $\begin{bmatrix} 5 & -2 \\ 9 & -6 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

5. $\begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$

6. $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$

7. $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

8. $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$

9. $\begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$

10. $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

11. $\begin{bmatrix} 6 & 2 & -2 \\ 2 & 5 & 0 \\ -2 & 0 & 7 \end{bmatrix}, \quad \lambda = 3$